

Towards a Model of Female Occupational Behavior: A Human Development Approach

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Substantial evidence suggests that women have added the role of worker to their characteristic adult roles of wife and mother. The inclusion of a work cycle in the female life course is a significant alteration of traditional roles, and is likely to have a major impact on both individual lives and the social structure. However, little has been done to incorporate these changes into theoretical perspectives on occupational development. This paper examines the literature on women and work by reviewing, at the societal level, the trends in female labor force activity since 1900 and, at the individual level, theories of the development of occupational behavior. An integration of these two areas is then suggested through an evaluation of their relevance to the human development paradigm.

Over the past quarter century, a dramatic change has occurred in the life course of American women through the addition of a work role to their traditional adult roles of wife and mother (Van Dusen & Sheldon, 1976). Even though this inclusion of a work cycle in the female life

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course will have a major impact on both the lives of individual women and on the social structure, only in the last decade has female labor force participation and its consequences attracted the attention of social and behavioral scientists. Existing research remains, however, limited in focus and scope by outdated empirical and theoretical conceptualizations of women and work. As Almquist (1977) notes, "most current research is macrosociological rather than microsociological, based on large-scale, impersonal, aggregated, and static data rather than small-scale, personal, disaggregated, and dynamic findings. From this we get a very firm appraisal of women's overall position in the labor force but we do not know the processes by which they attained it" (pp. 853-854). Consequently, although we do know about changes in female labor force participation over time, we know little about the developmental course of the work cycle in women's lives.

The task of explicating the work cycle of women clearly rests with social scientists of the life course, human developmentalists, whose perspective is uniquely suited to integrating and interpreting such changes in patterns of female labor force participation with changes in the development of the work cycle in women. However, in spite of the overwhelming evidence of life course changes in women's lives, little has been done to explain such changes in developmental terms or incorporate them into theories of the development of occupational behavior. In preparation for such a synthesis, this paper reviews existing literature related to women and work which includes, at the societal level, trends of female labor force activity from 1900 to 1975, and, at the individual level, theories of the development of occupational behavior. An integration of these levels of analysis will be suggested through an evaluation of their relevance to the human development paradigm. Thus, this paper attempts a multiple contribution by laying the groundwork for future studies of adult development to address the work cycle of women as a distinct process and as a component of human development, which should in turn stimulate research into a model of female occupational behavior.¹

¹In the literature, the behavior to which we are referring has many names: career development; occupational development; vocational behavior. We use the term "occupational behavior" in order to distinguish between the meanings of occupation and career. The term "career" has developmental implications in that a "career" normally has an orderly sequence of progressively higher-level stages within a small number of specific, elite occupations. Because it is clear that most women have "jobs" rather than "careers", we have chosen to refer to such activity as "occupational behavior" because all women who work exhibit occupational behavior but only a few exhibit career development.

TRENDS IN THE LABOR FORCE PARTICIPATION OF WOMEN, 1900-1975

At the turn of the century, there were three major occupations in which paid work outside the home was available to women: domestic service, factory work, and school teaching. The typical female worker at that time was young, unmarried, of lower-class or immigrant origins, and, unless in domestic service, white (Smuts, 1960). The primary socioeconomic determinants of female labor force participation were class, age and marital status. Paid employment was typically a prelude to women's real life work of bearing and raising families. When the family cycle began for women, the work cycle ended.

This pattern of alternating work and family cycles in women's lives remained stable between 1900 and 1940, although the percentage of young women working prior to marriage continued to increase (Oppenheimer, 1970, p. 7). However, during the post-war years, the timing of major events in the family cycle accelerated markedly (Glick, 1955, 1977), lengthening the post-parental stage when women are freed from the demands of intensive childcare.² Consequently, labor force participation by older, married women increased to such an extent that by 1960 the modal age of working women was 45-49 (Oppenheimer, 1970, p. 15).³ By the late 1960s, though, wives under age 35 were entering the labor force at a greater rate than older wives, which means that more women with children present in the home were working (Hayghe, 1976; Klein, 1975). More recently, the percentage of working mothers of pre-school-aged children has risen dramatically (Ferris, 1971; Nordheimer, 1977). Just as marriage proved to be no barrier to labor force participation by older women in the 1950s, the presence of children, even very

²To illustrate this change more specifically, the average mother in 1890 was 32 years old when she had her last child. She had borne 5.4 children within a period of ten years between marriage and the end of childbearing. Sixty years later, the average woman had borne her last child at age 26. The average number of children per woman was 2.35 and the interval between marriage and birth of the last child was about 6 years (Glick, 1955, p. 4).

³Oppenheimer's explanation of the phenomenon is a labor force *pull* of older women into the work force rather than a family *push* in that the major factors involved were an increased demand for *female* labor and the growth of certain occupations. Oppenheimer concludes that "the evidence indicates that the best explanation for the great increases in the work rates of older married women is that . . . in response to a declining supply of the young single woman, demand readjusted itself, so that workers who could be discriminated against were now acceptable, and perhaps even sought after" (Oppenheimer, 1970, pp. 188-189).

young children, in the home does not prevent large numbers of women from working today.

All these statistics indicate the aggregate of working women in 1975 bears little resemblance to the aggregate of working women in 1900. Marital status, family cycle stage, class and age are far less powerful predictors of female labor force participation today (Taeuber & Sweet, 1976). Clearly, the dominant position of the family cycle in the construction of the adult life course of women has been weakened. The lives of contemporary women do not exhibit the alternating patterns of work and family cycles characteristic of their mothers and grandmothers; rather, the evidence suggests that the female life course is now composed of work and family cycles in simultaneous operation throughout adulthood (Perun & Bielby, 1980).

The increase in female labor force participation in the course of this century can be accounted for not only by the changing rhythm of the family cycle but by changes in the post-war economy as well. The number of occupations which utilized female labor has increased greatly since 1900, and employment opportunities for middle-class women have expanded in white collar occupations such as clerical, service and sales work (Kreps, 1971; Wolf & Rosenfeld, 1978). The shift in the female labor force from blue-collar to white-collar jobs, however, has not lessened the extent to which the labor market is segregated by sex (Openheimer, 1970; Wolf & Rosenfeld, 1978). Women are predominantly employed in clerical, semi-professional, and service occupations which require fairly high levels of non-technical education and low levels of career commitment and which offer relatively high status and low pay. Even though changes in labor force participation patterns among contemporary women are linked to an increasing belief in less traditional sex-role attitudes (Mason, Czajka, & Arber, 1976), the occupational behavior of women in the aggregate is not linked to a reduction of sex segregation in the labor market. That is, although working women in 1975 were employed in different occupations from their counterparts in 1900, the variety of occupations open to them is similarly limited by sex segregation in the marketplace.

The steadiness of the increase in female labor force participation during the course of this century is impressive, occurring with little government encouragement or facilitation of women's attachment to the labor force, and there is little evidence to suggest its slackening or reversal in the future. Consequently, a re-analysis of earlier theorizing on the development of the work cycle of women is necessary if we are to understand the implications of these changes for both the lives of individual women and the social structure. This task calls for a theoretical perspec-

tive that synthesizes the outcome of a macro-level process with individual-level behavior. In the following sections the basis of such a perspective, that of human development, is described, and theories of the development of occupational behavior are discussed and evaluated.

LIFE-SPAN HUMAN DEVELOPMENT PERSPECTIVE

Human life-span development is a multi-disciplinary science drawn primarily from sociology and psychology whose primary concerns are the identification, description, and explication of ontogenetic (age-related) intraindividual behavioral change from birth to death and with the properties and characteristics of these changes and of the variables governing them (Baltes & Goulet, 1970). Human developmentalists view behavior as a dynamic process and address the complexities of that process through the study of the mechanisms, antecedents, consequences, and transitions that comprise behavioral development. Our use of the terms family cycle and work cycle therefore reflects a developmental orientation toward the study of major life roles. Human developmentalists are not restricted to the observation and specification of phenomena at the individual level of analysis, but are also able to identify and analyze the impact of social structure on the life course of the individual and vice versa. That is, human developmentalists are able to specify determinants of behavior as originating from both current and historical aspects of the social structure and the individual and can take into account the simultaneous effects of social structural and individual level factors.

Central to the human development paradigm is the conceptualization of behavior as the product of "a changing system of antecedent-consequent relationships operating over time" (Baltes, 1973, p. 460). Developmental researchers therefore focus on the analysis of cumulative, multivariate processes defined by time-related interactions between both individual and societal variables. Process is assumed to incorporate related behavioral responses which occur in an orderly sequence and group into phases or stages across the life span. Such stages become increasingly complex over time as more multivariate historical phases are antecedent to concurrent ones. Stages or phases occur continuously where there is little qualitative change and more quantitative change, and discontinuously when the reverse is true. The human development paradigm is, in the very broadest sense, an articulation of the interface of

all aspects of the individual with all relevant aspects of the social structure over time. Because it can account for this interface on both conceptual and empirical levels, the human development perspective is ideally suited for problems such as the specification of a model of vocational behavior.

Adaptation of the human development paradigm to the development of occupational behavior is relatively incomplete. Some vocational theorists (Cooley, 1964; Holland, 1963) have isolated important components of vocational behavior such as personality traits but have failed to conceptualize them as part of a complex process. Others (Lohnes, 1965; Mitchell, Jones, & Krumboltz, Note 1), have specified a process model in vocational development but have not described a comprehensive one. While Lohnes recognized process to be the central issue in occupational behavior, he did not complete the specification of the process by elaborating the presence of stages, the modes or mechanisms for change between stages, or the complexity of the process itself (i.e., possible interactions between historical, concurrent, individual, and social structural factors). Mitchell, Jones, & Krumboltz (Note 1) identify career decisions of individuals both as transition points and as "mechanisms" for change in the development of occupational behavior. However, they do not consider the decisions themselves to be qualitatively different across life or as life-stage markers. Among occupational theorists, examination of occupational behavior as an ontogenetic process over the life course is minimally considered. Finally, nearly all theorists who posit a process of occupational development focus solely on the individual level of analysis and rarely address the historical or social context of the process (Borow, 1964).

At present, research into the adult years among women concentrates on the family life cycle rather than on the work cycle, and no formal theory of the development of female occupational behavior exists. Early attempts to address the work cycle of women have not been successful either in generating research or in stimulating theoretical advances, primarily because of insufficient specification of a model (Psathas, 1968) and questionable assumptions about women and work (Zytowski, 1969). Within the area of vocational psychology, however, there are several formal theories relating to occupational behavior at the individual level which might contribute to an understanding of the work cycle of women. In the next section, such theories will be evaluated for their relevance to women and to the human development paradigm.⁴ A

⁴We are indebted in this paper to Samuel Osipow who has written an excellent and comprehensive review of theories of career development (see Samuel Osipow, *Theories of*

PSYCHOLOGY OF WOMEN QUARTERLY

<u>Criteria</u>	<u>Categories</u>			
	<u>Personality</u>	<u>Trait Factor</u>	<u>Vocational</u>	<u>Self-concept</u>
School of Thought	Personality theory (Psychology)	Personality theory (Psychology)	Vocational psychology (Psychology)	Self-concept theory (Psychology)
Focus	Satisfaction of needs	Match of interests and abilities with environmental opportunities	Vocational choice	Vocational development
Type of Theories	Semi-development	Non-developmental	Developmental	Developmental
Life Course Emphasis	Early childhood	None	Late adolescence to early adulthood	Childhood and early adulthood
Contributor	Roe	Holland	Ginzberg	Super

FIGURE 1. Theoretical approaches to the development of occupational behavior.

typology of the theories is presented in Figure 1. The evaluation criteria of these theories are found in the following questions:

1. Does the theory describe and explain a process?
2. Does the theory have a life course perspective?
3. Is the theory comprehensive? That is, can it incorporate contemporary social trends which have altered the traditional normative configuration of adult roles for women?

SELECTED THEORIES

Roe

The theory of vocational choice proposed by Anne Roe (1966) is based not on direct empirical research but on her synthesis of the work of personality and ego psychologists with Maslow's "need" theory and

(continued)

Career Development (2nd ed.), New York: Appleton-Century-Crofts, 1973). For a similar review in the area of occupational sociology but without reference to women, see Seymour Spilerman, "Careers, Labor Market Structure and Socioeconomic Achievement," *American Journal of Sociology*, 1977, 83, 551-593.

with genetic and evolutionary theory. The empirical catalyst to Roe's theoretical work was her study of the origins and development of personality differences among groups of eminent scientists (Roe, 1952, 1953). The results of those studies were then integrated with the above theories in Roe's formulation of the present theory of vocational choice.

A diagram of the major components of the theory is presented in Figure 2. The first grouping includes variables either given at birth (psychic energy and genetic factors) or developed early in life through parental interactions. Roe emphasizes the importance, particularly at the unconscious level, of innate talents, aptitudes and psychic energy as significant predispositions to vocational choice. Parenting style, however, is a more critical variable in that early parent-child interactions determine the child's "development of need primacies." According to Roe, the parents' attitudes toward the child largely shape the pattern and intensity of gratification of needs developed by an individual which are subsequently related to later levels of motivation and an ultimate career choice. That is, "needs satisfied routinely as they appear do not develop into unconscious motivators" (Roe, 1966, p. 112) but "needs, the satisfaction of which is delayed but eventually accomplished, will become unconscious motivators" (Roe, 1966, p. 113). Thus, the need hierarchy developed by an individual in response to a particular parenting style channels the choice of an appropriate vocation.

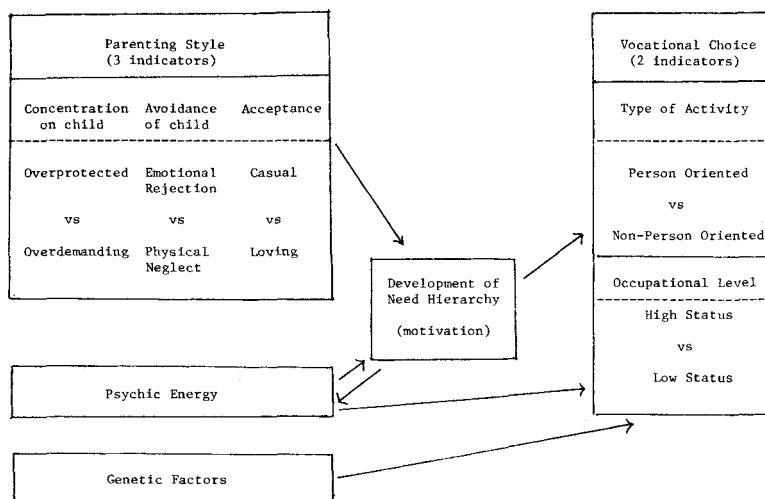


FIGURE 2. Roe's theory of vocational choice.

To summarize, Roe's theory of vocational choice states that the parenting style mediates the need hierarchy developed by an individual which in turn determines the ultimate vocational choice in the context of genetically determined skills and abilities.

Holland

Holland's (1963) theory of vocational choice is derived from the trait-factor tradition in personality research and represents an important extension of that tradition through its attempt to specify the means by which different personality types translate self-knowledge into vocational choice within a social context. Holland posits that most people organize their beliefs about the world of work through the use of occupational stereotypes whose study can reveal important information about personality dynamics in the same way that projective techniques give insights into unconscious motives. Therefore, the relationship between personality and vocational choice is mediated to a large extent by stereotypic typologies of occupations. In his research, Holland studied several large samples of students participating in the National Merit Scholarship program using a methodology of multiple observations of occupational behavior over specified time spans, some moderate and some long in length, as the empirical basis of his theory.

Holland organizes the critical elements in his theory according to hierarchical structures at both the individual and societal levels of analysis which explain major factors in the vocational choice process. The first is a classification of six major work environments in American society as motoric, intellectual, supportive, conforming, persuasive, or esthetic environments. At the individual level is a classification of six specific personality types corresponding to each occupational environment. In the motoric orientation would be found people who prefer dealing with concrete problems because they, in Holland's view, prefer to "act out" rather than "think through" problems. People in the intellectual orientation, in contrast, work independently with problems of an ambiguous nature. Those in the supportive orientation prefer work reflecting their desire for attention in a structured setting and their ability to solve problems through feelings and interpersonal relations. In the conforming orientation are people comfortable with structured and subordinate roles who achieve their goals by identifying with authority. People of the persuasive orientation, on the other hand, choose loosely-defined work situations where they can exercise verbal skills and acquire power, status, and leadership. Finally, those in the esthetic orien-

tation prefer to deal with objects in the environment rather than people. A concept Holland calls the level hierarchy determines the specific level within an occupation reached by an individual which can be predicted by the following formula: occupational level = intelligence + self-evaluation where self-evaluation is a function of socioeconomic characteristics, level of education, self concept, and drive for achievement. In a final hierarchical scheme, the developmental hierarchy, Holland posits an idiosyncratic ordering of preferences toward the major occupational environments by each individual. In Holland's view, the stability of the vocational choice is a function of such rank orderings of orientations because certain groupings are more suitable for some occupations than others.

There are three stages to the operation of Holland's vocational choice process. In the first, "a person directs himself toward the major occupational class for which his development has impelled him by selecting the occupational class at the head of his particular hierarchy of classes" (Holland, 1966, p. 132). Then, "within a major class of occupations, the person's selection of an occupation is a function of his self-evaluation and his ability (intelligence) to perform successfully in his chosen environment" (p. 133). Finally, both of these stages "are mediated by a series of personal factors, including self-knowledge and evaluation, knowledge of occupation classes . . . the orderliness of the developmental hierarchy; and a series of environmental factors . . . imposed by socioeconomic resources and the physical environment" (p. 133). Successful completion of the process by the individual depends on the soundness of the structure in the developmental hierarchy and the accuracy of the ranking in the level hierarchy.

To summarize, Holland has posited a theory of vocational choice in which an individual's knowledge of self and of the world of work interact to facilitate an occupational choice. Through a complex ranking procedure of many dimensions, the individual undergoes a process by which personality dispositions are fitted to a specific occupational setting within a framework of certain abilities, skills, and socioeconomic factors.

Ginzberg

In the early 1950s, Eli Ginzberg and his colleagues presented a general theory of occupational choice based on the following premise:

Our basic assumption was that an individual reaches his ultimate decision, not at any single moment in time, but through a series of

decisions over a period of years. . . . We sought to delineate the process, discern the most significant patterns, and describe the major deviations. (Ginzberg, 1966, p. 95)

By the emphasis on maturation and the specification of the process of occupational choice, Ginzberg was clearly concerned with the explication of a developmental phenomenon.

The basic propositions in this theory of occupational choice, which is depicted in Figure 3, are that occupational choice is a process; that the process is largely irreversible; and, that compromise is an essential aspect of every choice. The second element of the theory, the irreversibility of the choice process, contains the contribution of the social structure to the process at the individual level. In many instances, the social structure precludes alterations in career choice later in life. The third element, that occupational choice is a compromise, also refers to the interaction of societal and individual factors. To make an appropriate choice, the individual must be aware of environmental constraints and opportunities. The empirical foundation of the theory is a study of an elite sample of 60

STAGE Substage	AGE RANGES	DEVELOPMENTAL TASK	GOAL OF STAGE
FANTASY	0-10		transition from play to work orientation
TENTATIVE:			development of sense of skills and interests
Interest	11-12	identification of activities liked or disliked	
Capacity	13-14	consideration of abilities in addition to interests	
Value	15-16	awareness of social needs and personal values	
Transition	17-18	formation of practical and immediate training plans	
REALISTIC:	19-22		selection of appropriate occupation
Exploration		selection of major field of study	
Crystallization		focusing on a specific career choice	
Specification		selection of specific career or commencement of specialized training	

FIGURE 3. Ginzberg's theory of the development of occupational choice.

boys who ranged in age from 11 to 24. In order to test the generalizability of the theory, a sample of lower-class boys and a sample of upper-class women were also studied. From the results of this research, Ginzberg concluded that the theory could be fruitfully applied to these different samples even though significant differences among them were found.

Subsequently, Ginzberg (1972) suggested a major conceptual shift by proposing a life-span perspective in place of the earlier emphasis on adolescence and early adulthood. He also limited the concept of irreversibility by suggesting that a choice could be revised but at great cost to the individual. Finally, the concept of compromise is now one of optimization defined as the individual's potential to experience the occupational choice process throughout the life-span in a continuing search for the best fit between abilities, interests, and available opportunities.

To summarize, Ginzberg and his associates have formulated a theory of occupational choice which describes a process generally occurring during adolescence in which specific developmental tasks related to vocational maturity are sequentially mastered. By the early twenties, the successful completion of these tasks results in a specific vocational choice. An appropriate vocational choice is a reflection of successful progress through the stages of vocational development.

Super

Ginzberg's theory of vocational choice was immediately challenged by Super (1966) who criticized its failure to incorporate earlier research in vocational psychology into his theoretical framework. A more telling criticism was the lack of specification of the means by which the final stage of compromise between environmental and individual factors is reached. In Super's view, that compromise "is the crux of the problem of occupational choice and adjustment: the nature of the compromise between self and reality, the degree to which and the conditions under which one yields to the other, and the way in which the compromise is effected" (Super, 1966, p. 102). Super's subsequent integration of biological, psychological and sociological influences related to vocational choice has resulted in an inclusive theory of vocational development throughout the life-span. The empirical test of Super's theory was a longitudinal study begun in 1951 with a study sample to be followed over a twenty-year period of 142 ninth grade boys in a school in upper New York State.

STAGE Substage	AGE RANGES	GOAL OF STAGE
GROWTH	1-14	growth of self-concept through interaction with family and peers
EXPLORATION: Tentative Transition Trial (little commitment)	15-22	testing of vocational choices through school, part-time work and play
ESTABLISHMENT: Trial (commitment) Advancement	23-44	implementation of a career choice and stabilization within an occupation
MAINTENANCE	45-64	continuity of vocational behavior along established lines
DECLINE	65+	preparation for retirement and relinquishment of work role

FIGURE 4. Super's self-concept theory of the development of vocational behavior.

The major propositions of this theory of vocational choice are: First, that people differ in personality, abilities, and interests and are therefore qualified for a variety of occupations, while each occupation requires a certain pattern of personality traits, abilities and interests in its workers. Second, people change with time and experience so that vocational choice and subsequent adjustment to that choice is a continuous process. Finally, the process of vocational development is essentially that of developing and implementing a self concept through a complex, interactional process of genetic aptitudes, physiological make-up, opportunity to play social roles and self-evaluation. That process is charted in the series of vocational stages depicted in Figure 4.⁵ Further, self-concept, the personality component of the theory, is a continually evolving construct which becomes increasingly differentiated and complex with maturation. Therefore, over time a vocational self-concept, a sub-system of the general self-concept, emerges. Although Super does not specify its

⁵Super has also developed an elaborate set of vocational tasks which is too lengthy to be included here. The interested reader is directed to Super, Starishevsky, Matlin and Jordaan (1963).

developmental process, the vocational self-concept is presumed to evolve through the mechanisms of identity-testing, identification, and interaction with the environment during the early life stages critical to vocational development.

To summarize, Super has posited a multi-level, multi-stage theory of vocational development. Organized within a system of life-stages, each requiring different vocational behaviors, the theory states that vocational development is a process by which the individual reaches a compromise between abilities and interests and the demands of the environment. Through the development of a vocational self-concept and the mastery of vocational tasks, the individual successfully completes the stages of preference, choice, entry, and adjustment in the occupational world.

Evaluation of Theories

Both Roe and Holland fail to meet one or more of the human development criteria of process, comprehensiveness and a life course perspective. While Roe identified sources of stages in a process, her static model does not explain how the variables interact after early childhood. Possible dialectical relationships between parent and child or inconsistencies in parenting style are never considered. Nor does Roe ever address the social context in which her model is embedded. While Holland does credit the individual's perception of the social structure with an important role, his theory has a major limitation in his uncritical use of occupational stereotypes as a key theoretical concept. Vetter (1975), among others, has discussed how such stereotypes cause employers to exclude qualified women, and women to exclude themselves, from many occupations. Holland's work is also flawed by his failure to specify a developmental process. The origins of traits important to vocational choice are never identified, and there is no sense of the time in which the fit between individual and occupational hierarchies should occur. Thus, neither Roe nor Holland presents theoretical formulations adequate for a developmental approach to the occupational behavior of women.

Ginzberg and Super do satisfy the stipulated developmental criteria. However, Ginzberg's use of an elite sample of college women, whose work cycles are probably not representative of typical employment patterns, to test his theory raises serious questions about its general validity. Consequently, a realistic specification of the work cycle of women, including the effect of interruptions in work histories or education (Tittle

& Denker, 1977) on stage completion or subsequent stages, cannot be determined through Ginzberg's theory. Although Super's theory has not been tested on a female sample, its incorporation of personal and situational determinants in a life course framework comes closest to satisfying fully our developmental criteria. However, the focus of Super's theory remains microsocial, thus excluding important social, structural and economic factors, and recent extensions of the theory (Putnam & Hansen, 1972) indicate that the self-concept, an integral part of the theory, is implemented differently in women. Because we cannot assume that the process of self-concept development and its contingent stages is identical for both sexes, further empirical work is needed before Super's theory of the development of occupational behavior can be considered applicable to women.

CONCLUSION

It is clear that either major revision of existing theory or formulation of new theoretical perspectives is necessary for the development of a model of female occupational behavior. As Leona Tyler has expressed it, "Much of what we know about the stages through which an individual passes as he [sic] prepares to find his [sic] place in the world of work might appropriately be labelled 'The Vocational Development of Middle Class Males'" (Tyler, 1977, p. 40). While this situation has been recognized by vocational theorists, proposed paradigm revisions (Psathas, 1968; Zytowski, 1969) remain virtually unused (Tittle & Denker, 1977; Vetter, 1975). We posit, therefore, that a solution to the current theoretical morass will be found only when women's occupational behavior is viewed from the human development paradigm which emphasizes process, comprehensiveness, and a life course perspective. We find an increasing sophistication of developmental concepts applied to the study of vocational behavior in the reviewed theories but with obvious limitations. Failure to consider social structural constraints on vocational choice, failure to explain more than vocational choice (even when a more complete developmental sequence is posited), and failure to consider changes in vocational choice later in life or to extend the critical period of choice beyond late adolescence—all these limitations argue for an investment in the creation of new theories, rather than a rehabilitation of the old.

The contributions of the human development perspective to our understanding of the work cycle of women derive from its ability to

integrate macrolevel and microlevel phenomena. In contrast to the reviewed theories, the human development paradigm recognizes that social structure and social change have a profound impact on the occupational behavior of women. Given the extent and persistence of sex segregation in the labor force, the process of vocational choice among women is constricted by social and economic factors perpetuating occupational sex stereotyping. Analysis of the antecedent-consequent relationships which constitute the work cycle of women therefore must take into account the concentration of women in a small number of occupations similarity limited in terms of skill requirements, opportunities for advancement, and status. The assumption of existing theories that vocational choice is a function of inter-individual differences does not hold in the case of women. Incorporation of structural characteristics of the labor market into theoretical formulations is therefore imperative to the study of the work cycle of women.

Although our criticism of the theories reviewed above has been based on their inability to explain gender differences in the development of occupational behavior, the human development perspective permits us to take our analysis one step further. Given the temporal foundation of the human development paradigm, it is apparent to developmental researchers that existing theories are flawed not only by their sex-inappropriate content but by their age-inappropriate structures as well. That is, it is the interaction of age and gender roles, rather than gender roles alone, which can account for the occupational behavior of contemporary women. As we have seen, current theories explain vocational development within the context of the adult male life course in which the work cycle predominates. In contrast, the adult female life course has traditionally been constructed in accordance with the events of the family cycle. Only recently have women added a work cycle to their adult lives, and, as a result, the timing and sequence of events in women's lives have been fundamentally altered (Perun & Bielby, 1980).

These changes in the adult life course patterns of contemporary women have profound implications for theories of the development of occupational behavior. Specifically, they suggest that the determinants of occupational behavior of women are different from those of men, that the trajectory of the work cycle of women is less predictable than that of men, and that the process of synchronizing work and family cycles throughout adulthood may be more difficult for women than for men. For example, particularly in their 20s, women may be faced with asynchronous cycles in that major events in the biological cycle (the prime period for childbearing) and the family cycle (marriage and childbearing) can occur simultaneously with important events in the work cycle

(specialized training or intensive career-building activity). Because the timing schedules in each of these cycles are set, more or less implicitly, by reference to norms of age-appropriate behavior which are established by the social structure (Neugarten & Datan, 1973) and which differ for men and for women, new theories of the development of occupational behavior of women which can account for the temporal complexities of the female life course must be forthcoming, given the changes in women's life patterns during the course of this century. Thus, the timing of events in the work cycle of women raises new issues in vocational theory. These issues must be examined both within the context of the work cycle itself and in relation to other developmental cycles in the adult life course, if we are to understand how age and sex interact to determine the occupational behavior of women.

In conclusion, the specification of the work cycle of women is a complex task in which the biological, social, historical, and psychological processes simultaneously present in the adult life course must be considered within a context of major changes in the life course patterning of women. The developmental criteria of process, comprehensiveness, and a life course perspective not only suggest directions for theory creation but provide the bases necessary for the specification of a model of female occupational behavior as well. Hence, the human development perspective makes a valuable contribution through its potential for integrating multiple factors into systematic, coherent description and explication of the work cycle of women, which ultimately should result in a model of female occupational behavior.

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